Yixuan Wang

Ph.D. Student, University of Florida · Expected May 2027 · Gainesville, FL

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RESEARCH PROFILE

Machine learning researcher working at the intersection of nonlinear control and modern ML. I study learning and generative algorithms as dynamical systems and turn their stability and control structure into new algorithms with rigorous guarantees. First-author publications at ICML (Spotlight), IEEE CDC, and IFAC World Congress; co-author at NeurIPS. Seeking research scientist positions beginning summer 2027.

SELECTED RESEARCH CONTRIBUTIONS

- **Effective Model Pruning (ICML 2026 Spotlight).** Introduced a method that converts any importance-score distribution over model components into an adaptive sparsity level via effective sample size, with a provable bound on the induced loss change; implemented and validated across MLPs, CNNs, Transformers/LLMs, and KANs.
- **Riemannian Lyapunov Optimizer.** Formulated optimization as a controlled dynamical system on a Riemannian manifold; the Lyapunov construction recovers classical optimizers and systematically generates new update laws; implemented and benchmarked on ImageNet-scale classification with ResNet and ViT architectures.
- **Generative modeling as distributional control.** Developed a Lyapunov perspective on energy-based generative models in which sampling is a control problem, yielding a principled finite-step stopping rule for Langevin sampling.
- **Bayesian intent inference and privacy.** Built a Rao-Blackwellized particle filter for online goal inference from trajectories, and control policies that jointly regulate physical motion and an observer's belief, with formal bounds on goal information disclosure (IFAC 2026, IEEE CDC 2026).

TECHNICAL SKILLS

Machine Learning	PyTorch; generative models (energy-based, diffusion, flow matching); reinforcement learning; model compression
ML Systems	Multi-GPU distributed training (DDP); Slurm-based HPC workflows (UF HiPerGator)
Programming	Python, C/C++, MATLAB / Simulink, Git, LaTeX

SELECTED PUBLICATIONS

1. Effective Model Pruning: Measuring the Redundancy of Model Components — *ICML 2026* Spotlight
2. Off-Dynamics Reinforcement Learning via Domain Adaptation and Reward Augmented Imitation — *NeurIPS 2024*
3. Quantifying Trade-Offs Between Stability and Goal-Obfuscation — *IEEE CDC 2026*
4. Goal Inference with Rao-Blackwellized Particle Filters — *IFAC World Congress 2026*
5. Riemannian Lyapunov Optimizer: A Unified Framework for Optimization — *arXiv preprint*

EDUCATION

University of Florida — Ph.D., Mechanical & Aerospace Engineering	2024 – 2027 (exp.)
Johns Hopkins University — M.S., Computer Science	2023 – 2024
University of Florida — M.S., Mechanical Engineering	2020 – 2022
Tongji University — B.S., Mechanical Design, Manufacture and Automation	2015 – 2020